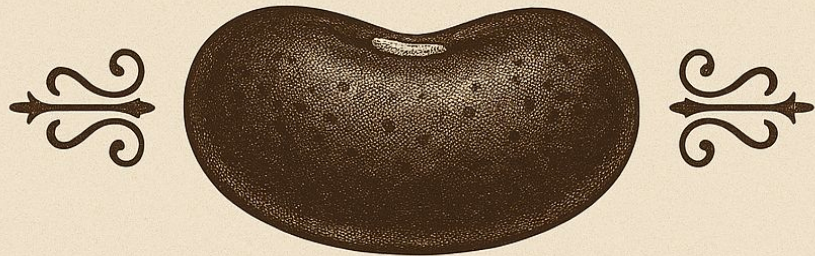

BIG BEAN DEALER



LLC

America's Leading Wholesale
Field Bean Distributor

PROVIDING 70% OF
U.S. & 45% OF GLOBAL
BEAN SUPPLIES

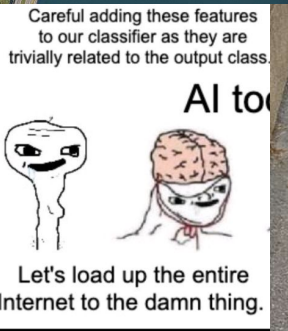
LARGEST SELECTION
OF BEANS
WORLDWIDE

PREMIUM QUALITY & FLAVOR

Meet Our Team....

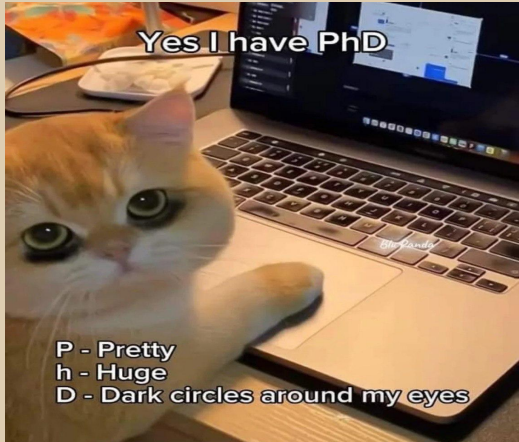


David "Bean Lord" Jean CEO



Osman “Data Lama” Ali

COO

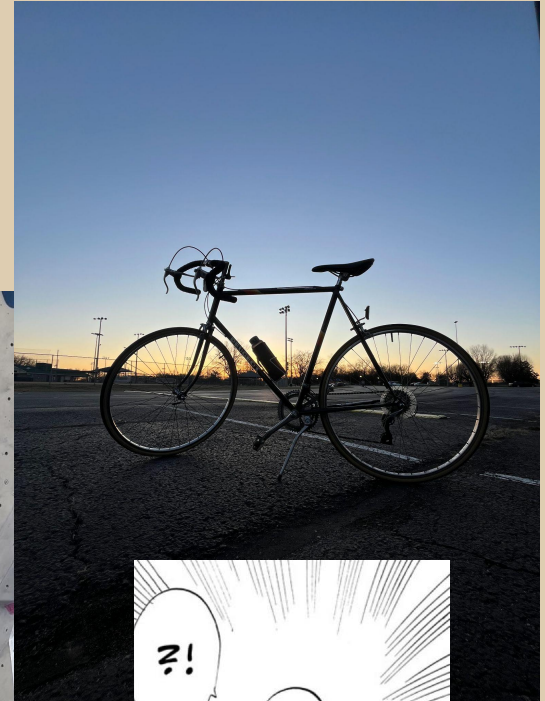
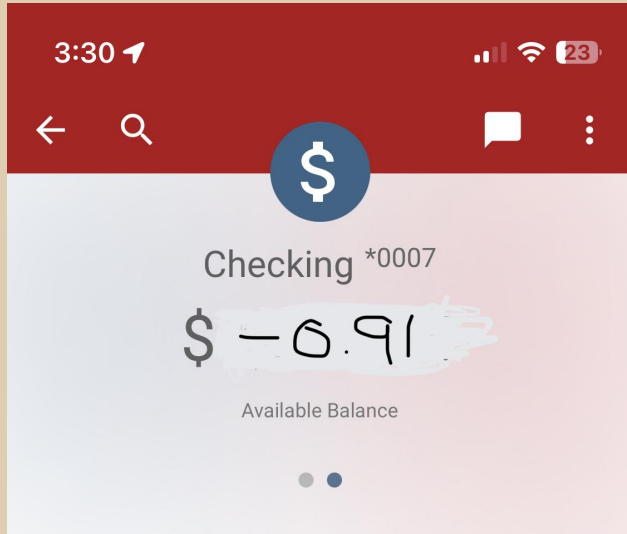


$$\frac{\partial}{\partial \theta} \log \pi(\xi) = \frac{\partial}{\partial \theta} \int_{\mathcal{R}_n} T(x) f(x, \theta) dx = \int_{\mathcal{R}_n} \frac{\partial}{\partial \theta} T(x) f(x, \theta) dx$$
$$\frac{\partial}{\partial a} \ln f_{a, \sigma^2}(\xi_1) = \frac{(\xi_1 - a)}{\sigma^2} f_{a, \sigma^2}(\xi_1) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\xi_1 - a)^2}{2\sigma^2}\right)$$
$$\int_{\mathcal{R}_n} T(x) \cdot \frac{\partial}{\partial \theta} f(x, \theta) dx = M\left(T(\xi) \cdot \frac{\partial}{\partial \theta} \ln L(\xi, \theta)\right) = \int_{\mathcal{R}_n} T(x) \left(\frac{\partial}{\partial \theta} \ln L(x, \theta)\right) \cdot f(x, \theta) dx = \int_{\mathcal{R}_n} T(x) \left(\frac{\partial}{\partial \theta} \frac{f(x, \theta)}{f(x, \theta)}\right) f(x, \theta) dx$$
$$\frac{\partial}{\partial \theta} \log \pi(\xi) = \frac{\partial}{\partial \theta} \int_{\mathcal{R}_n} T(x) f(x, \theta) dx = \int_{\mathcal{R}_n} \frac{\partial}{\partial \theta} T(x) f(x, \theta) dx = \int_{\mathcal{R}_n} T(x) \left(\frac{\partial}{\partial \theta} f(x, \theta)\right) dx$$



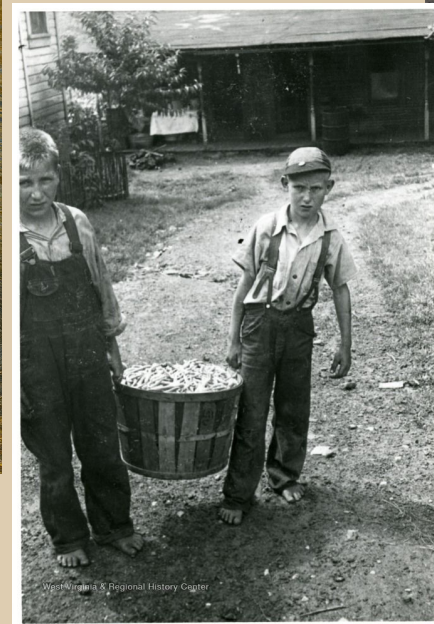
Stiven “Free Solo” Lavrenov CTO

“I’m like ride bike
And I like climb rock”



Company History

From humble beginnings...



Company History(cont)

...To a global powerhouse

Leading the world in:

logistics, robotics AI, & ML..and of course

... BEANS 



More about or tech and models

AI-POWERED OPTIMIZATION FOR BEAN OPS ©

QUALITY CONTROL (COMPUTER VISION)

Tools: OpenCV + YOLOv8
/EfficientNet

Usage: Detects defects in
beans during sorting
using image
classification

ROUTE OPTIMIZATION

Tools: Deep Reinforcement
Learning (DRL), GNNs

Usage: Adjusts delivery
routes in real time
based on:

- Traffic Warehouse
- Weather load
- Fuel prices
- Warehouse load

VRP Solver

SALES

Tools: XGBoost/
LSTM/Prophet

Usage: Forecasts bean
demand by
region and time
using historical
sales + external
factors (e.g.
seasonality,
region trends)

Company Values

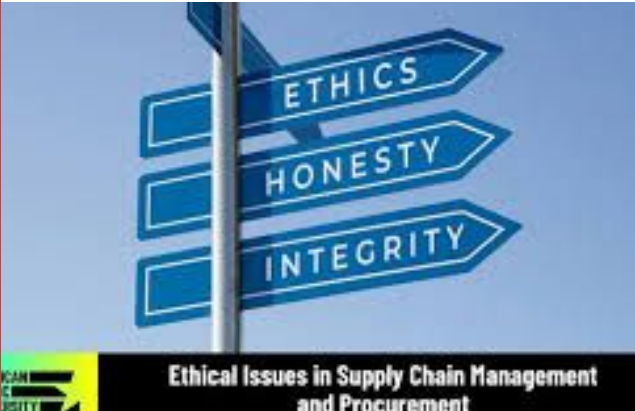


EQUITY
Ensuring everyone has the same opportunities, which often means treating each

DIVERSITY
Constantly and consistently recognising and valuing differences

INCLUSION
Creating a culture and environment where everyone feels safe to be true their self

BELONGING
Everyone feeling valued, accepted and celebrated for who they are



Ethical Issues in Supply Chain Management and Procurement

...And for that reason our customers love and support us



SHOP

WWW.BIGBEANDEALERLLC.COM

**Hoodies
only
\$80 USD**



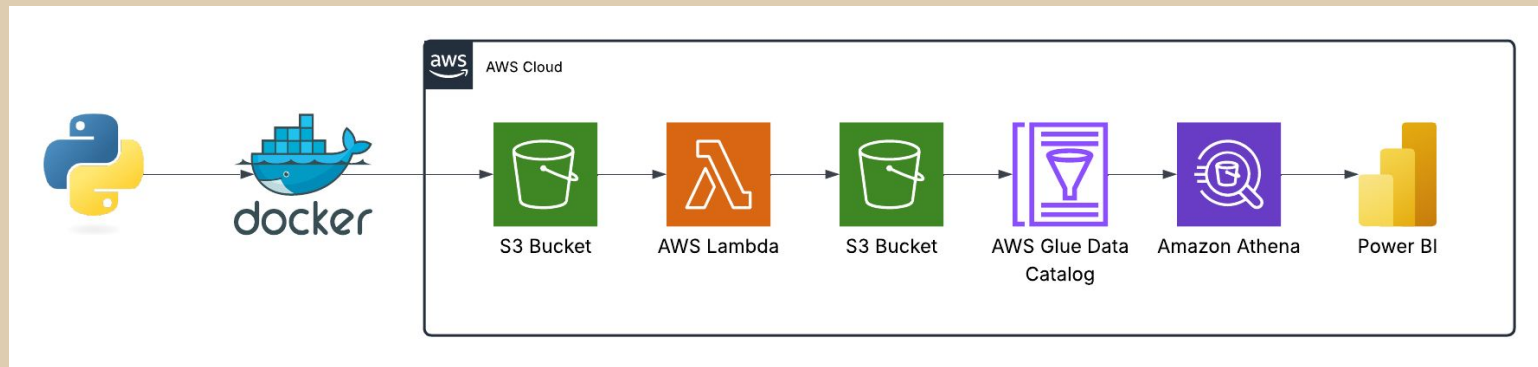
**Get your
Premium
Beanware
today!**

Despite all our tech products...

Today, in front of a group of young tech savvy entrepreneurs, we are here to talk to you about the number 1 way our company makes and saves money.

And that is robust and efficient data infrastructure

One example of that is our **serverless ETL data pipeline with event driven architecture**



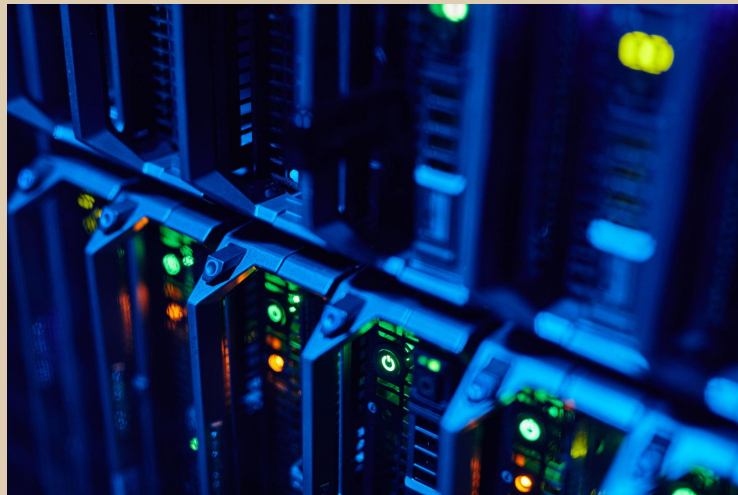
Problem Statement

Key Challenges:

- Messy, incomplete, and unstructured data hinders decision-making.
- Manual processing is time-consuming and error-prone.
- Lack of automation leads to operational inefficiencies.
- Dedicated servers cost a lot of money

Solution:

- Implement a **serverless** ETL pipeline on AWS.
- Enable real-time access to structured, high-quality data.



Project Objectives

- Automate sales data ingestion, cleaning, transformation, and storage.
- Reduce manual intervention while ensuring accuracy.
- Provide **scalable, cost-effective, and serverless** data infrastructure.
- Improve business intelligence through **real-time querying and analytics**.



Expected Insights & Business Impact

- **Sales Performance Trends:**
Revenue growth tracking.
- **Product Procurement Optimization:**
Prevent stock outs & overstocking.
- **Operational Efficiency:** Improve fulfillment rates, reduce costs.
- **Competitive Edge:** Real-time data-driven decision-making.



Methodology

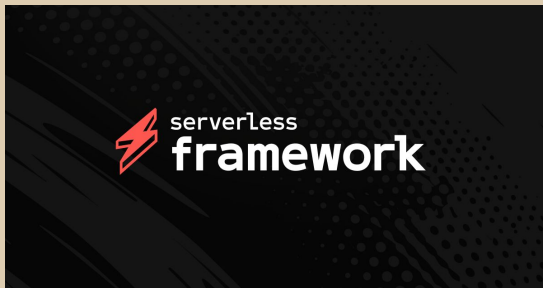
oa

Analytical Techniques & Methodology

- **Focus:** Exploratory Data Analysis (EDA), not ML.
- **Key Techniques:**
 - **Statistical Aggregations:** Sales trends by time period.
 - **Comparative Analysis:** Sales breakdown by region, category, etc.
 - **Visualization:** Power BI dashboards with time series, bar graphs, heatmaps.
- **Performance Metrics:**
 - Data accuracy, completeness, and consistency.
 - Revenue growth, seasonal trends, customer segmentation.



Frameworks Used



DJ

Terraform – Infrastructure as Code (IaC)

- **Manages state:** Tracks infrastructure in a local or remote `terraform.tfstate` file, ensuring consistent and idempotent deployments.
- **Declarative syntax:** Uses HCL (HashiCorp Configuration Language) to define cloud resources, which makes infrastructure predictable and version-controlled.
- **Supports modularity:** Encourages reusable, composable modules for complex infrastructure setups.
- **Cloud-agnostic:** Works across AWS, Azure, GCP, and more — great for multi-cloud or hybrid environments.
- **Lifecycle automation:** Handles `create`, `update`, and `destroy` workflows with dependency resolution between resources.

Serverless Framework – Application Deployment for Serverless Architectures

- **Simplifies serverless deployment:** Automates packaging and deployment of AWS Lambda functions, API Gateway routes, DynamoDB tables, etc.
- **Built-in monitoring and rollback:** With `serverless` dashboard or plugins, you get deployment tracking, rollback support, and alerts.
- **Config-driven:** Define infrastructure and function logic in one `serverless.yml` file, reducing boilerplate.
- **Plugins and extensibility:** Huge ecosystem of plugins for offline testing, parameter store injection, and more.
- **Works with Terraform:** Can coexist with Terraform by excluding infrastructure components that Terraform already provisions (e.g., IAM roles or S3).

Great Expectations – Data Validation and Testing Framework

- **Expectation suites:** Define reusable tests (expectations) for datasets, such as null checks, ranges, regex patterns, and unique constraints.
- **Data context:** Manages configurations, expectations, and results — stored locally or in cloud (e.g., S3, Databricks, etc).
- **Validation checkpoints:** Run validations manually or in automated pipelines (CI/CD, Airflow, Lambda, etc).
- **Rich documentation:** Generates visual reports (“Data Docs”) showing validation success/failure and metadata about datasets.
- **Seamless integration:** Can run inside Spark, Pandas, or SQL workflows and integrates well with modern data stacks (Airflow, Prefect, dbt, etc).

AWS Services

OA



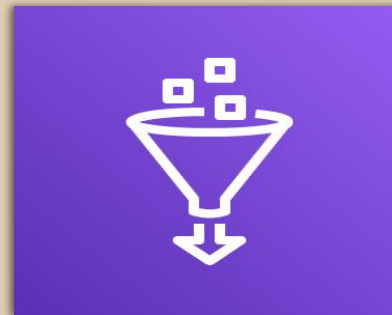
AWS S3 Bucket

- Object storage service for data lake architecture.
- Stores raw, curated, and processed data.
- Easily integrated with Glue, Athena, and Lambda.
- Offers high availability, versioning, and lifecycle management.



AWS Lambda

- Serverless compute service to run code in response to events.
- Triggers data processing when new files are added to S3.
- Scales automatically, no need to manage infrastructure.
- Supports Python, Node.js, Java, and more



AWS Glue Catalog

- Central metadata repository for datasets.
- Automatically catalogs and tracks schema changes.
- Enables data discovery and schema evolution.
- Integrates with Spark, Athena, and ETL jobs.



AWS Athena

- Serverless interactive query service.
- Uses standard SQL to query data directly from S3.
- Fast, cost-effective analytics without needing to set up a database.
- Ideal for log analysis, ad hoc queries, and dashboarding.

Make

Build pipeline components locally with reproducibility, containerization, and version control to prepare for cloud deployment

Run Locally

- Simulates daily sales data for testing
- Calls `process_sales_function()` from `run_local.py`
- Cleans and structures data for downstream use
- Validates pipeline logic without AWS costs

Create Requirements

- Extracts dependencies from Poetry environment
- Generates `requirements.txt` for deployment
- Ensures Lambda runtime has needed packages
- Keeps builds reproducible and lightweight

Setup Docker

- Prepares Docker image for Lambda packaging
- Runs `docker-build` with correct dependencies
- Matches AWS Lambda's runtime environment
- Ensures consistency across local and cloud

Version Control

- All code, infrastructure, and pipeline steps tracked in GitHub
- Enables structured collaboration and code reviews
- Supports reproducible builds across environments
- Centralized source of truth for all deployments

Deployment

Deploys the packaged Lambda function to AWS using Serverless Framework, which registers all resources via CloudFormation for automated, consistent, and scalable infrastructure

Build Lambda Package

- Creates deployment-ready zip with code and deps
- Uses Docker to match AWS Lambda runtime
- Ensures compatibility for cloud execution
- Clean, reproducible packaging process

Serverless Deploy

- Runs `sls deploy --stage development`
- Uploads Lambda + metadata to AWS
- Registers handler, environment, and runtime
- Deploys to appropriate stage (e.g., dev/prod)

CloudFormation Stack

- Serverless auto-generates CloudFormation template
- Provisions Lambda, IAM roles, and S3 permissions from Terraform
- Tracks all resources as a named stack
- Supports rollback and full-stack deletion

Lambda Registration

- Lambda function is deployed and named in AWS
- Linked to triggers (like S3 or EventBridge)
- Becomes invocable by other AWS services
- Lives in AWS Lambda console for validation/debugging

Everything Put Together:

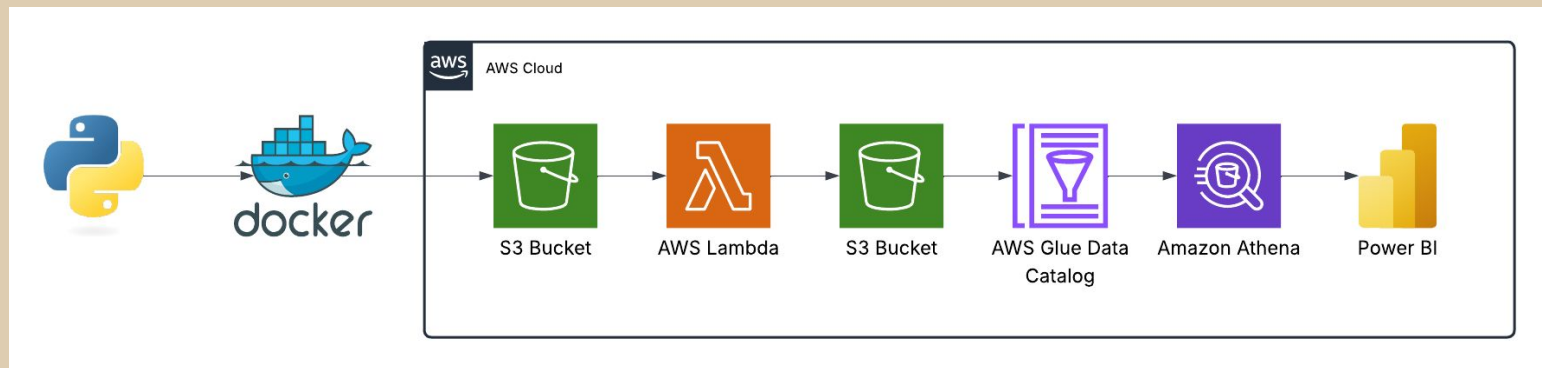
Our serverless ETL data pipeline with event driven architecture

Architecture Diagram

- **Visual Representation of Data Flow:**
 - Sales data ingested into S3.
 - AWS Lambda triggers data transformation.
 - Processed data stored in S3 & registered in Glue Catalog.
 - Athena enables fast, serverless querying.
 - Power BI used for visualization and reporting.

Infrastructure as Code & Deployment Strategy

- **Terraform for Infrastructure Management**
 - Automated provisioning of S3, IAM, Glue, Athena.
 - Ensures consistency across environments.
- **Serverless Framework for Lambda & Scheduling**
 - Automates deployment and job execution via EventBridge.
- **GitHub for Version Control**
 - Repository management, structured collaboration, and controlled deployments.



Our Data

Dataset Identification

- **Source:** Self-generated using the Faker library.
- **Format:** Parquet (optimized for querying in Athena).
- **Size:** Daily ~60 KB as Parquet, ~146 KB as CSV
- **Key Attributes:**
 - Customer Name, Customer Address, Bean Type, Amount Purchased, Cost, Sales, Profit, State, Sale Date, Customer Region, etc.
- **Data Processing Steps:**
 - Handling missing values.
 - Data type enforcement.
 - Deduplication.
 - Dimensional modeling for efficient querying.

sales_data	
customer	NVARCHAR
address	NVARCHAR
bean_type	NVARCHAR
amount_purchased_lbs	INT
purchase_cost	DECIMAL(8, 2)
sell_price	DECIMAL(8, 2)
profit	DECIMAL(8, 2)
state	VARCHAR(2)
region	NVARCHAR
date	DATE

Dashboard Demo

How Much Does This Cost?

dj

Build your own pipeline!!!

Here is a starter repo

- Great for file structure
- Can be used for DE or ML OPS

